Recursive self-improvement for long-running agents

The problem it solves, how it differs from giving an agent a memory, and what the measurements show so far

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Abstract

An agent that runs for months acts on knowledge that goes stale, and nobody tells it. A model, a prompt full of its own history, or a vector store can tell the agent what it did before. None can tell it that this has stopped working, write the corrected rule, prove it on the failing cases, release it to a share of traffic and take it back if it does harm. That loop is what we mean by recursive self-improvement. SenseLab's engine implements it in four levels, and this paper reports what it measured on a preregistered benchmark in which the correct action for a task class changes part way through, for half of the class only. On four held-out seeds, two agent models and two task families (1,600 episodes per system), the engine takes the new action first in 45 % of the episodes whose rule moved, against 25 % for the model alone, 32–36 % for three pgvector variants and 40 % for a 24k-token context window; it wins 8 to 11 of 16 paired cells against each, no pairing survives Holm correction, and the gain is concentrated in `support`. It holds the classes that did not change at 100 % against 77 % for the model alone, at a third of the cost of the context window; on the kept half, the rule that did not move within a changed class, the context window is ahead of the engine (92 % against 69 %). On development seeds, a joining agent was right on 122 of 122 first exposures (strongest retrieval baseline 68–75 %), and in an exploratory run the gate shipped four corrections and stopped a wrong one at 88 % recurrence against 10 %. Inside a 100-episode cell the gates add nothing measurable to the engine's first level, outcome attribution; that level is more than a win/loss count, since the engine beats a DIY outcome counter by 13 points on the moved half and 16 on the unchanged classes. The four levels are one loop; where we know which level carried a number, we say so.

1 The problem

Most evaluations of agents measure one task at a time. A model reads a task, acts, and is scored. Deployment is different in one way that matters more than any other: the agent keeps going. A support agent handles tickets for a year. A CI agent repairs builds for every commit a team pushes. An operations agent runs the same procedure every night. Over that time the world the agent works in changes, and almost none of the changes arrive as instructions.

Three properties of this setting decide what kind of system can work in it.

- *Tasks do not repeat.* Every ticket, build failure and incident is a new instance. Nothing can be memorised verbatim; what carries over is the rule, "in this situation, take this action".
- *Rules change silently and partially.* A policy is rewritten for one tier of customer and kept for the rest. A dependency is pinned for one class of package and left alone for others. The agent's old rule is now right half of the time, which is the hardest case: a system that only demotes what failed has no signal, because the rule still succeeds often enough to look fine.

- *The cost of a wrong rule compounds.* An agent that acts on a stale rule fails on every later instance of that situation, and so does every other agent that reads the same rule. A fleet of ten agents sharing one record repeats one mistake ten times per round.

The failure that matters is a confident action on what used to be true, repeated until a person notices. The question for anyone building agents for long-running work is how the system itself can notice, correct and verify, without a person in every loop, and without making things worse when its correction is wrong.

2 Three ways to keep an agent current, and where each stops

The approaches in use today put the burden in three different places.

Trust the model. The agent runs on a frontier model with a fixed prompt. Everything it knows about the operation is what the model learned in pretraining plus what the prompt says. This is the baseline every deployment starts from. It has one strength: the model's default policy is usually the textbook one, and when the operation follows the textbook the agent is right. It has one weakness that the benchmark below makes concrete: when the operation's rule differs from the textbook, the model has no way to learn that. In our confirmatory run, the frontier model alone takes the correct post-change action first in 25 % of the episodes whose rule moved and in 27 % of the episodes in the changed classes whose rule did not move. In the `support` family its moved-half rate is 0 of 150. It never learns the house rule because it has nowhere to write it.

Put the record in the prompt. The agent is given its own history: every prior note and outcome from the operation, most recent first, as many as fit. The model then reasons over the record. This works well on one thing. When a rule is non-standard and has not changed, the record is full of consistent examples and the model follows them: 92 % on the kept half in our run, the best of any system by a wide margin. It fails on the thing this paper is about. When a rule has changed, the record contradicts itself, old successes against new failures, and the model has to work out from raw history which half is current. It does so in 40 % of moved-half episodes. The approach also has two structural limits. The record grows without bound and must be truncated (24k tokens here), so the oldest evidence is the first to go, and every episode pays to re-read everything: $0.130 per episode, three times the cost of the engine and nine times the cost of the model alone. And it does not transfer: what one agent has in its prompt is invisible to the next agent until it is copied there.

Retrieve from a vector store. The agent writes notes after each task, embeds them, and retrieves the nearest ones before the next task. This is what most "agent memory" systems are [3, 4, 6], with variations: a cross-encoder reranker to sharpen retrieval, or a per-entry win/loss counter so the agent can see how often a note led to success. We ran all three. Plain pgvector retrieval takes the new action first in 32 % of moved-half episodes; with a reranker, 36 %; with the win/loss counter, 32 %. All three keep an unchanged non-standard rule at 58–66 %, well below the context window's 92 %. The reason is in what retrieval answers. A vector store answers "what did I write that looks like this?". It does not answer "is what I wrote still true?". A reranker makes the first answer sharper and leaves the second unasked. A win/loss counter is a step toward the second answer, and in our run it did not help on the moved half and cost 12 points on the unchanged classes (84 % against 96 % for plain retrieval). A counter on an entry says how often it worked. It

does not say when, or what to do instead. Hosted memory products were not run here; an earlier grid on the same benchmark family ran Mem0 [2].

The common thread is that all three treat knowledge as something to store and recall. What they leave out is a process: something that watches whether recalled knowledge still works, writes the corrected rule when it does not, tests the correction, and releases it carefully. That process is the subject of the next section.

3 What recursive self-improvement means here

We use the term in a narrow, operational sense. A system is recursively self-improving when its own outcomes change the knowledge it acts on, the changes are themselves tested against outcomes before they take effect, and the machinery that makes the changes is subject to the same evidence as the knowledge it changes. The word "recursive" points at the last clause: the corrections apply to the thing that produces corrections, and the record the loop keeps is also what trains its next version.

The term is also used for systems that rewrite their own code or training toward open-ended capability gain [9, 10]. Nothing here does that. The engine changes the operating knowledge its agents act on, and the prompts and models that produce corrections to it; every change is tested against outcomes it did not write, exposed to a share of live traffic before all of it, reversible, and logged. The gate is the design.

SenseLab's engine implements this as four levels. Everything below is production code; the benchmark calls the public API and the engine's own scheduler does the rest.

Level 0: outcomes decide what the agent acts on. Before acting, an agent asks the engine for a briefing and a retrieval on its situation. As it works it records the actions it takes and any attempt that failed. When the task ends it commits the outcome. The sealed trace carries exactly what was read, and the outcome is attributed to it: an entry read by a run that then failed is charged; an entry read by a clean success is credited. Every entry carries an evidence status (untested, validated, contested, discredited), and retrieval returns, with each hit, how each candidate action has fared here on similar tasks and a recommendation: act, explore, or escalate. The "explore" recommendation is assigned per agent, so ten agents facing the same uncertainty try different things instead of the same one.

This level already differs from a vector store in kind. The store answers by similarity. Level 0 answers by evidence attached to the similar thing, and that evidence comes from outcomes the engine observed, including the agent's own failed attempts. A rule that was validated three times and then fails is flagged as a *regime shift*: the world moved under a rule that used to be right. The engine also scopes the shift to a *situation*, a canonical label the agent attaches to each session, so a rule that broke for the paid tier is not demoted for the free tier.

Level 1: the engine proposes, tests and ships corrections. A regime shift proposes a fix automatically. SenseLab's repair agent, a platform-side component of the engine, reads the failing runs, the entry and the loop's history on that entry, and drafts either a corrected entry or a conditional procedure (conditions, steps, a verify step) on a branch of the shared record. It attaches a manifest: the runs it predicts it repairs, the runs it must not break, and an expected prevention rate. Nothing on the branch is visible to agents yet.

The draft then has to pass three gates.

1. *Stand-in replay.* A stand-in model re-runs each manifest case with the branch as its memory and the tool calls answered from the recorded run. A replay that could not be answered from the record (fidelity below 0.5) does not count as a pass.
2. *The customer's own replay.* The case set is delivered to the customer's agent, which re-runs it on the branch. The fix must turn the failed runs into successes and must not break the passed ones. A success reached only after a recorded failed attempt fails the case.
3. *A live canary.* Sessions are bucketed by a hash of (fix, session); a share reads from the branch, the rest from the main record, and writes always go to main. The canary is the harm gate [12]: the replays prove the benefit on the runs that failed, and the canary has to show that the change does not make the agent worse on the live traffic the old rule still served. It passes when the canary arm recurs significantly less than control, or when a one-sided 95 % bound on the difference in recurrence is under a 10-point margin while the failure is still live in control. It fails when the canary arm recurs significantly more.

A fix that passes is merged into the main record. Its prevention is verified over the following window and a fix that regresses is rolled back to the pre-fix versions. A policy can instead ship on a passed replay, skipping the canary; that is the "− canary" ablation below. A set of continuous judges grades sealed traffic and opens incidents, the second source of proposals after regime shifts.

Level 2: the loop reads its own history. Every event in a fix's life (proposed, tested, replayed, shipped, regressed, rolled back) is written back as evidence on the entry it concerned, and the repair agent's next draft on that entry reads it. A loop that proposes the same wrong fix twice has not learned; this level is where it can.

Level 3: the record trains the next model and the next improver. The traces the loop produces, including the repair trajectories and the cases they pinned, are a training set. Smaller models are distilled from them and served with the same evidence record as their memory. The repair prompt itself is A/B tested across proposals and scored by proven fixes.

Why the gate is the point. A model that drafts fixes from failure evidence will draft wrong ones. In the exploratory run reported below, one draft said that a timed-out export should be split into date ranges; the rule had moved to a settings change, and the draft would have failed 88 % of the sessions it touched. A system that writes such a draft directly into what every agent reads damages every session at once. The discipline that makes self-improvement usable in production is the order of operations: evidence before proposal, proof before release, a small exposure before full exposure, reversibility after, and an audit trail throughout. Generating corrections is the easy part; the model does that. The engine's value is in refusing most of them, and in keeping a record of what it refused and why.

4 What the measurements show

The numbers in this section come from a preregistered study of the whole engine [1]. The hypotheses, arms, metrics, seeds and analysis were registered before the first run; two seeds were used for development and four held-out seeds ran once after the analysis code was frozen. This paper quotes the confirmatory read, and labels the two development-seed probes and the one exploratory run as such.

The analysis was frozen; the engine was not. Defects in the loop's evidence path that the study found were fixed in production between runs, and every run carries the engine cut it ran on. Table 1 is the fleet-1 confirmatory read: sixteen cells per system, run 01:42–08:19Z on 2026-09-30. One engine cut fell inside that window (`post-canary-zero`, 03:40Z), with 4,867 of the 12,791 episodes before it. No SenseLab cell straddles the cut: the `support` cells on GPT-5.5 ran on the earlier cut, and every `ci-fix` cell and every Claude Opus 5 cell ran on the later one. Because no cell ran on both, the two engine versions cannot be compared on the same cells; what can be said is which cells ran on which. On the earlier-cut cells alone the moved-half difference between SenseLab and every other system is +28/+11/+6/+18/+23 points, against +19/+4/+13/+9/+13 on the full read; the two differ in composition, since the earlier-cut cells are the `support`/GPT-5.5 slice, where the model alone is at 0 % and the context window at 17 %, and the full read adds `ci-fix`, where the context window leads. The ten-agent cells of Figure 2 and the exploratory run both ran on the later `post-replay-evidence` cut, after the last defect was fixed. The cut of every run is listed in the companion paper's reproducibility section [1].

The benchmark. Two task families with tool-using agents and oracle-graded outcomes: `support` (a helpdesk agent with account, order and export tools; 15 issue classes) and `ci-fix` (an agent repairing failing CI on a repository; 10 failure classes). Every task instance is generated and never repeated, with random identifiers so nothing can be read off an id. Each cell is one agent identity working 100 episodes with one store seeded with 300 distractor entries. At episode 30 the correct action for a subset of classes changes *conditionally*: it moves for instances with one attribute value (the *moved half*) and stays for the others (the *kept half*). Half of the post-change tasks come from the changed classes. Classes that did not change are the third population (*unchanged*). The design was chosen because on an unconditional change outcome learning alone recovers nearly everything; the conditional change is the case where demotion has no signal, because the old rule is still right half of the time.

Six systems ran on identical prompts and the same two agent models (GPT-5.5 and Claude Opus 5): the frontier model alone; the context window (every prior note and outcome, most recent first, 24k tokens); pgvector (bge-small embeddings, cosine top-5 over notes); pgvector with the cross-encoder reranker SenseLab serves; pgvector with notes, periodic LLM consolidation and a per-entry win/loss counter shown to the agent (the "DIY outcomes" arm); and SenseLab, the whole engine with corrections shipped only after a passed canary. Four seeds, two families and two models give sixteen cells per system, 1,600 episodes each. The primary endpoint is first-attempt success on the moved half after the change; the secondary endpoints are first-attempt success on the kept half and on the unchanged classes.

4.1 After the rule changes

Table 1 and Figure 1 carry the main result. Before the change every system with a store is at 76–86 % and the model alone at 65 %: a store of any kind teaches the house rules. After the change the systems separate on the population the change created. On the moved half SenseLab takes the new action first in 45 % of episodes (127/283, cell-level interval [39, 51]); the context window is at 40 %, the pgvector arms at 32–36 %, the model alone at 25 %. On the unchanged classes SenseLab is at 100 % (556/557) and no system is above it. On the kept half the context window is ahead (92 % against 69 %), and the engine is level with the pgvector arms.

Paired on (family, seed, model) over the sixteen cells, SenseLab's moved-half rate is above every other system: +19 points over the model alone, +4 over the context window, +13 over pgvector, +9 over pgvector with the reranker and +13 over the DIY outcome store, winning 8 to 11 of the 16

Table 1: Confirmatory read, sixteen cells per system (4 seeds × 2 families × 2 models), 1,600 episodes per system. First-attempt success before the change and, after it, on the three populations; episode success counts an episode solved within the retry budget of three; escalated is the share of episodes the agent handed off; cost is the all-in dollar cost per episode, including for SenseLab the loop's judges, replays and drafts.

system	pre-change	moved half	kept half	unchanged	episode	escalated	$/ep.
Frontier alone	65 %	25 % (72/284)	27 % (75/276)	77 % (432/560)	85.1 %	14.9 %	0.014
Context window (24k)	86 %	40 % (115/284)	92 % (253/276)	97 % (542/560)	92.2 %	7.7 %	0.130
pgvector	82 %	32 % (91/284)	66 % (183/276)	96 % (539/560)	91.8 %	8.2 %	0.024
pgvector + rerank	77 %	36 % (101/284)	66 % (183/276)	91 % (511/560)	89.7 %	10.3 %	0.027
pgvector + outcomes (DIY)	76 %	32 % (90/284)	58 % (161/276)	84 % (470/560)	88.2 %	11.8 %	0.032
SenseLab	85 %	45 % (127/283)	69 % (191/276)	100 % (556/557)	95.2 %	5.0 %	0.042

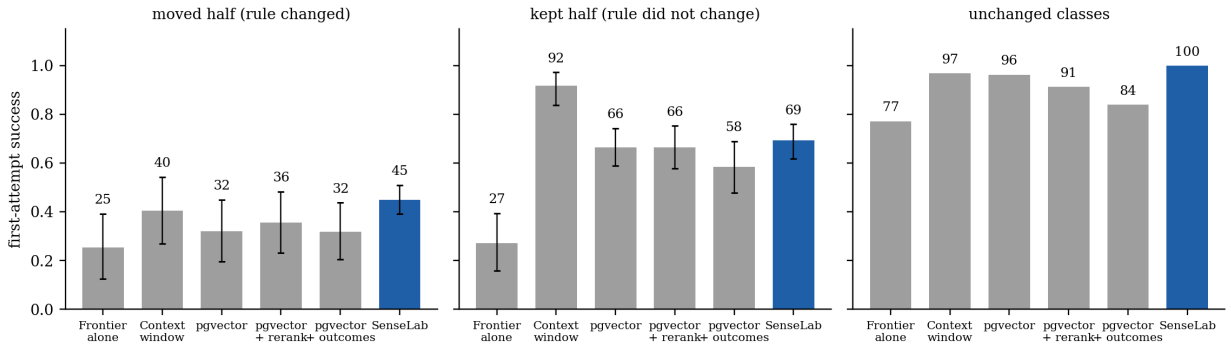


Figure 1: The three populations of Table 1. Bars are first-attempt success after the change; whiskers on the moved and kept halves are cell-level bootstrap intervals over the sixteen cells.

cells in each pairing. The exact sign-flip test gives $p = 0.024$ against the model alone, 0.040 against pgvector and the DIY store, 0.119 against the reranker and 0.544 against the context window; after Holm correction across the five comparisons none is below 0.05 (0.121, 0.544, 0.159, 0.239, 0.159). The sign is right in every pairing and the sizes match the development seeds; sixteen cells is the power this study bought. The reader should take the deltas as estimates with their intervals, and the context-window comparison on the moved half as undecided.

The secondary endpoints are cleaner. On the unchanged classes SenseLab is above the model alone (+23, Holm 0.002), the reranker (+9, 0.035) and the DIY store (+16, 0.004), and indistinguishable from the context window and plain pgvector. On the kept half it is above the model alone by 38 points (14 of 16 cells, Holm 0.007), level with the three pgvector arms, and 24 points below the context window (1 of 16 cells, Holm 0.001). That last number is a real cost of the approach and we return to it in §7.

Where the gap is large. The moved half is the primary endpoint because it is the hardest population, and there the engine's lead is modest. On the other endpoints the lead is not modest, and Table 2 collects the ones a reader deciding what to deploy should weigh. One miss in 557 on the classes whose rule did not change: 100 % against 77 % for the model alone, 84 % for the DIY outcome store and 91 % for the reranker, all three separations surviving Holm. Forty-two points over the model alone on the `support` moved half, where every system without an outcome record is at 27 % or less and the model alone never takes the new action. A third fewer escalations than the next-best system and a third as many as the model alone: 80 hand-offs in 1,592 episodes

Table 2: Where the confirmatory gaps are large. Rates over the sixteen cells; Holm-corrected p from the preregistered paired sign-flip test where the comparison is one of the registered secondary endpoints, blank otherwise.

endpoint	SenseLab	against	gap	Holm p
unchanged classes, first attempt	100 % (556/557)	Frontier alone 77 %	+23 pts	0.002
		pgvector + outcomes 84 %	+16 pts	0.004
		pgvector + rerank 91 %	+9 pts	0.035
kept half, first attempt	69 % (191/276)	Frontier alone 27 %	+38 pts	0.007
`support` moved half	42 % (62/149)	Frontier alone 0 % (0/150)	+42 pts	
		best other (rerank) 27 %	+15 pts	
moved half, Claude Opus 5	54 % (76/141)	Context window 45 %	+9 pts	
pre-change first attempt	85 % (406/480)	Frontier alone 65 %	+20 pts	
episode success (retry budget 3)	95.2 %	Context window 92.2 %	+3 pts	
		Frontier alone 85.1 %	+10 pts	
escalations	5.0 % (80)	Context window 7.7 % (124)	−35 %	
		Frontier alone 14.9 % (238)	−66 %	
cost per episode, all in	$0.042	Context window $0.130	3.1× cheaper	

against 124 for the context window and 238 for the model alone. The highest episode success of any system, 95.2 %, with a retry budget of three. Twenty points over the model alone before any rule changes, because the engine's record teaches the house rules an agent is deployed into. And on Claude Opus 5 the moved-half lead over the context window is nine points, where on GPT-5.5 the two are level.

Where the gain comes from. The moved-half result is a `support` result. There SenseLab is at 42 % (62/149) against 0–27 % for every other system, with the model alone at 0 of 150: on house rules that differ from the textbook and then move, nothing without an outcome record learns the new rule at all. On `ci-fix` SenseLab is at 49 % (65/134) against 43–67 %, behind the context window (67 %). The two families differ in what the textbook gets right: in `ci-fix` the frontier model alone is at 54 % on the moved half, which suggests the post-change action is close to its default, and at 5 % (7/146) on the kept half, where a house rule it never learned still applies. By model, SenseLab is level with the context window on GPT-5.5 (36 % both) and ahead on Claude Opus 5 (54 % against 45 %); the pgvector arms swap order between the models (13–27 % on GPT-5.5, 37–51 % on Opus). The engine's effect does not depend on which model reads it.

4.2 The long horizon: ten agents, 200 episodes

A 100-episode cell with one agent is short for a loop whose gates need graded live traffic. The second protocol runs ten agents against one shared engine for 200 episodes, with the change at episode 50, on the same four held-out seeds, both families, GPT-5.5 (eight cells, 1,600 episodes). It is reported descriptively, as registered, and it has no baseline: no pgvector or context-window fleet ran under this protocol, so the curve compares the engine with its own earlier episodes, and a shared store of any kind might climb as evidence accumulates. What the protocol adds that the single-agent cells cannot is the gate's behaviour with ten agents' traffic, reported next.

Figure 2 shows the curve. Pooled over the eight cells, moved-half first-attempt success is 37 % (40/109) in episodes 50–99, 55 % (60/110) in 100–149 and 75 % (67/89) in 150–199, with 88 % (15/17) in the last block. The kept half is 66, 76 and 72 % over the same windows. The unchanged

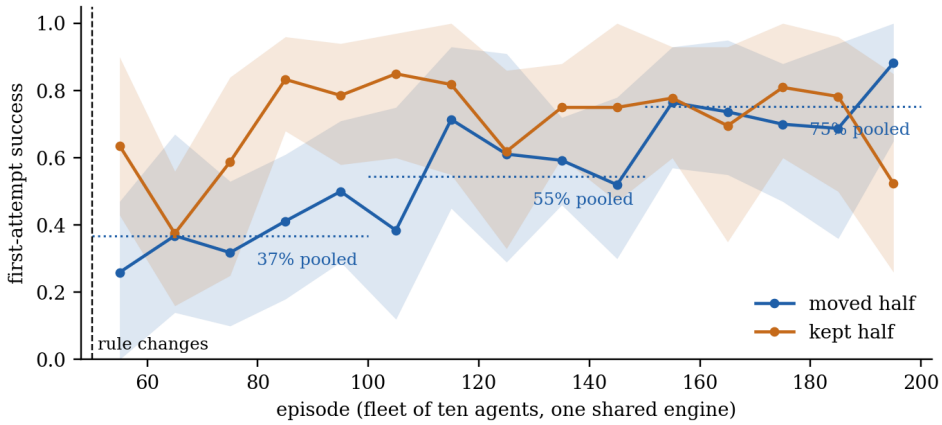


Figure 2: Fleet of ten agents sharing one engine, 200 episodes, change at episode 50; eight confirmatory cells pooled. First-attempt success per ten-episode block on the moved and kept halves, with cell-level bootstrap bands; dotted lines are the pooled moved-half rate over each fifty-episode window. No baseline system ran under this protocol; the curve is descriptive.

classes are at 100 % in every post-change block (594/595). A rule that moved for half of a class at episode 50 is taken correctly three quarters of the time by episode 150, by ten agents each of which handled about a tenth of the episodes. What each inherits from the others is the outcome record and whatever the loop wrote.

4.3 What the loop did, and what it refused

The engine's own record is the second half of the evidence, and it is where the gate shows.

In the sixteen single-agent cells the loop proposed 32 corrections (23 from a regime shift, 9 from the judges). Sixteen passed the stand-in replay, six passed the customer's replay, five opened a canary, two shipped and none was reverted. The same loop with the canary removed, so that a passed replay ships, shipped 10 of 44. The canary held back four fixes in five that the replay would have released, in cells where no canary could reach a graded decision before the cell ended. In the eight ten-agent cells the loop proposed 48, the customer's replay passed 6 and failed 38, six canaries opened in six different cells, and none shipped: four reached the graded floor on both arms, one failed on a zero-count bound at 10 of 10 graded sessions, and three closed with both arms above the floor and no decision, because at 14 to 20 graded runs per arm the non-inferiority test cannot close a 10-point margin. The canary arm read its fix in 57 of 76 graded sessions (75 %) and the control arm in 0 of 54, so the gate was grading a treated arm against an untreated one, as designed.

The exploratory run sized the traffic to the gate. On the two development seeds, four cells of 400 episodes with every post-change episode drawn from the changed classes, the loop proposed 68 corrections; the stand-in replay passed 20, the customer's replay passed 8 and failed 22; eight canaries opened and five decided inside their cells. Four passed and shipped, three of them on a `ci-fix` class the fleet already handled (first-attempt success 0.89–0.97 before the ships, 0.89–0.96 after), so what the gate certified there is that the correction did no harm. One failed. The draft "csv export timeout → split date ranges" was the wrong action under the changed rule, which had moved to a settings change. Its canary arm graded 8 sessions, read the fix in 7, and recurred in 7 (88 %) against 2 of 20 in control (10 %; $p < 0.001$). The loop wrote the failure as a lesson to the main record within a second, the branch never reached main, and the next draft the repair agent produced for that class named the settings action. Nobody injected this case. It came out of the

loop's own proposals, and the gate caught it.

What did not separate. Inside a 100-episode cell with one agent, the engine without its canary and the engine without its repair loop are indistinguishable from the whole engine: 44 / 71 / 99 % and 43 / 68 / 99 % (moved / kept / unchanged) against 45 / 69 / 100. Level 0, the outcome record, does most of the measurable work at this horizon. The repair loop's contribution is visible in its lifecycle counts and in the exploratory canary, and in the shape of the ten-agent curve; it is not visible in the single-agent aggregate. We say so because the case for the loop has to rest on what it prevents as much as on what it adds, and the former is only measurable where it has traffic.

Level 0 is not a win/loss counter. The DIY outcome arm is the naive version of it: notes, periodic consolidation, and a per-entry success count shown to the agent. The engine beats that arm by 13 points on the moved half (10 of 16 cells, $p = 0.040$ before correction), by 16 on the unchanged classes (Holm 0.004) and by 8 on the kept half, and escalates less than half as often (5.0 % against 11.8 %); the ablation with the repair loop switched off, which is level 0 by itself, is at 43 / 68 / 99 % against the DIY arm's 32 / 58 / 84. The difference is what happens between an outcome and the next retrieval. The engine attributes each outcome to the exact entries the session read and the attempts it recorded, so a failure charges the rule that was followed and not every note that happened to be retrieved. It keeps an evidence status per entry, with a threshold for discrediting and a regime-shift rule (validated at least three times, now failing) that distinguishes a rule the world moved from a rule that never worked. It scopes all of that to a situation, so a rule broken for one tier is not demoted for another. And it returns a recommendation with the hits, act, explore or escalate, with exploration assigned per agent so a fleet does not repeat one agent's mistake. That machinery is the loop's first level, and the levels above it exist for the case it cannot see: a rule still right half of the time, which demotion cannot separate from a rule that works. Which level carried each number is stated wherever we can tell.

4.4 When the team grows

No operation runs one agent for long. A second instance is added for load. Another team points its own agent at the same tools and the same customers. A new use case is built over the same systems, by people who were not there when the first agent learned the house rules. Every one of these agents arrives with the same question on its first day: of everything written down about this operation, what still holds? With a vector store the answer is whatever the retriever returns, and the new agent has no way to tell a note that was right last quarter from one that is right now. With the engine the answer is attached to each entry, because the entries carry the outcomes of every agent that acted on them.

A development-seed probe measured what a joining agent inherits. Six agents share one record on the `support` and `ci-fix` families, two seeds, with an unconditional rule change at episode 20. In the *pioneer* protocol one agent works alone for 60 episodes and five join; in the *newcomer* protocol five work from the start and a sixth joins at 60. The joiner is a new identity; it has none of the veterans' sessions. Table 3 is the read.

A joiner on the engine took the post-change action on its first exposure to a changed class in 36 of 36 and 10 of 10 cases, and the correct action on every first exposure of any kind, 122 of 122, with no stale picks and no escalations. Over all of its episodes after joining it was at 100 % first-attempt success (136/136 and 160/160). A joiner on pgvector with a reranker was at 31 % and 40 % on the changed classes, 68 % and 75 % over all first exposures, and 62 % over all of its episodes, with 17 % stale picks in the pioneer protocol and 16–20 % of first exposures escalated. The record each joiner read explains the difference. The engine's joiner had a validated entry in context on every

Table 3: A joining agent's first exposure to each task class after it joins (development seeds, set A, six agents; pgvector + rerank is the strongest retrieval baseline). *Poisoned*: a failed first attempt with an untested peer note in context and nothing validated. *Parity*: the joiner's success reaches the veterans' late rate. Tokens are per task the joiner solved.

protocol	system	first exposure		poisoned	escal.	parity	ep. to	tokens /
		changed	all			reached	parity	solved
pioneer	SenseLab	36/36 (100 %)	90/90 (100 %)	0/90 (0 %)	0 %	20/20	5	4,331
	pgvector + rerank	11/36 (31 %)	61/90 (68 %)	29/90 (32 %)	20 %	16/20	5	7,556
newcomer	SenseLab	10/10 (100 %)	32/32 (100 %)	0/32 (0 %)	0 %	4/4	5	4,230
	pgvector + rerank	4/10 (40 %)	24/32 (75 %)	8/32 (25 %)	16 %	4/4	11.5	6,836

first exposure and was never poisoned. The retrieval joiner had no evidence status on anything it read, and on 32 % and 25 % of its first exposures it failed with an untested note from a veteran in its context; on the changed classes, 69 % and 60 %. The same notes were in both stores. What the engine's joiner inherited was the part the store cannot hold: which notes had worked since they were written. Onboarding cost follows: the engine's joiners all reached the veterans' late success rate, in a median of five episodes in both protocols; four of the twenty retrieval joiners in the pioneer protocol never did, and in the newcomer protocol the median was 11.5 episodes and 55k tokens against 22k. Per task solved, the engine's joiner spent 43 % and 38 % fewer tokens.

This is a development probe on an unconditional change with six agents and one hundred episodes, and we report it as such. Its direction is the one the organisation case needs. The cost of learning a changed rule is paid once, by whichever agent meets it first, and every agent that joins afterwards, on the same team or another, starts from the validated record instead of from the veterans' unverified notes.

4.5 Models trained on the record

The record a team's agents produce is also a training set, and this is level 3 of the engine. SenseLab builds *Managed Models*: smaller models that SenseLab fine-tunes on a team's own sealed traces, what the agents read, what they did, what came of it, and the repair trajectories and pinned cases the loop produced along the way. The trained model is then served with the same evidence record as its memory. Three things follow for the team. It owns the intelligence: the record was produced by its agents on its operation, and the model trained on that record belongs to it, on no vendor's roadmap. It can retrain as the record grows, so a rule the fleet learned this month is in the next model, and a rule the loop corrected is in it too. And the model it runs on is a fraction of the cost of the frontier model it learned from.

The first rung of this was measured on the development seeds. Two `senselab-s1` students were trained on 200 sealed `support` traces each from the development cells, one on the whole engine's traces and one on traces from the ablation with the repair loop switched off, and evaluated on three further held-out seeds (1,500 episodes) in three conditions: the student alone, the student served with the engine's record, and GPT-5.5 served with the same record. Table 4 is the read.

Alone, the students have learned the house rules and nothing usable about the change: 88 % and 100 % on the kept half, 9 % and 0 % on the moved half. (The engine-trained student alone does reach the right answer on 96 % of moved-half episodes, in 2.1 attempts on average: it has learned its teacher's recovery path, the stale action first and then the one the environment accepted, and replays it.) A fine-tuned model is a snapshot of the record at training time, and a snapshot cannot follow a rule that moves after it was taken. Served with the engine's record, the same students take

Table 4: Managed Models at the first rung of the evaluation ladder (development seeds 11 and 23 for training; seeds 101, 113 and 127 for evaluation; `support`, set C, 300 episodes per condition and model). First-attempt success by population, post-change episodes pooled over the three seeds ($n = 54$ moved, 51 kept, 105 unchanged, 90 pre-change per row); cost is agent cost per episode.

model	memory	moved half	kept half	unchanged	pre-change	$/episode
student (engine traces)	none	9 %	88 %	69 %	80 %	0.0007
student (ablation traces)	none	0 %	100 %	90 %	94 %	0.0007
student (engine traces)	SenseLab	33 %	69 %	98 %	71 %	0.0015
student (ablation traces)	SenseLab	30 %	65 %	97 %	84 %	0.0017
GPT-5.5	SenseLab	22 %	69 %	99 %	80 %	0.0348

the new action first in 33 % and 30 % of moved-half episodes, 24 and 30 points above themselves alone in every seed, and hold the unchanged classes at 98 % and 97 %. Against GPT-5.5 served with the same record they are 8 to 11 points ahead on the moved half, level on the kept half (69 against 69) and on the unchanged classes (98 against 99), at $0.0015 against $0.0348 per episode: 1/23 of the cost. The record is what moves with the world; the model is what makes acting on it cheap. The record was produced by one model's outcomes and lifted another, which is the property an organisation needs if it is to change models without starting over.

Two disclosures belong next to these numbers. This is rung 0 of the ladder: the evaluation seeds are new but the task classes and wordings are the ones the students trained on, so a student ahead of the frontier model here is expected and is not yet evidence of generalisation; rungs 1–3, on classes and wordings the students never saw, are registered and not yet run. And both students were promoted to serving under a relaxed gate; the default promotion gate would have rejected both on exact agreement with their teacher (0.00 against a floor of 0.45). The difference between the two students, $-7/0/+18$ points across the three seeds, is small and reported as such. What the rung does show is the shape of the offer: a model the team owns, trained on what its agents learned, that with the record behind it matches a frontier model on the house rules and the unchanged work and beats it where the rule moved, at a twenty-third of the cost.

5 What this means for building agents

The measurements support six statements about how to build agents for long-running work.

The unit of improvement is the operating rule. What goes stale is a rule of the form "in this situation, take this action". Retraining the model for every policy change is too slow and too coarse; editing the prompt by hand does not scale past a few agents. The rule is the thing to store, attribute outcomes to, correct, and test. Everything in Table 1 that separates the systems is about how each one handles that unit.

Evidence attached to knowledge beats more knowledge. The context window has strictly more information than the engine: it sees everything. It loses on the moved half because information without attribution is a contradiction the model has to resolve on every call. The engine's record is smaller and carries, with each rule, how it has fared since. On the kept half, where there is no contradiction, more information wins, and this is the engine's clearest weakness in the data.

The gate matters more than the generator. The repair agent drafted 68 corrections in the exploratory run and 32 and 48 in the confirmatory protocols. The customer's replay rejected most of them. The canary rejected one that the replay had passed and that would have harmed nine sessions in ten. A system whose corrections are gated by outcomes it did not write itself can be left running. One that ships on a model's say-so has to be watched.

An organisation learns once. Agents multiply: more instances on the same team, other teams on the same tools, new use cases over the same systems. A new agent on the engine took the correct action on every first exposure measured, 122 of 122, where the strongest retrieval baseline failed on a quarter to a third of them with a veteran's unverified note in its context. The curve in Figure 2 is ten agents reaching 75 % on a rule that most of them met for the first time after it had already been learned. The cost of learning a changed rule is paid once per operation, by whichever agent meets it first.

The team owns the intelligence, and can run it on its own model. The record is produced by the team's agents on the team's operation, and the Managed Models SenseLab trains from it are the team's own, retrainable as the record grows. At the first rung measured, a student served with the record was level with a frontier model on the house rules and the unchanged work, ahead where the rule moved, and 23 times cheaper. Accuracy comes from the record, which keeps moving with the world; cost comes from the model, which the team controls.

Self-diagnosis is an output. The study that produced these numbers found fourteen things about the engine's own evidence path, twelve of them defects, during the runs: a regime shift firing on entries that had never been validated, a replay grading on an outcome the agent reached only after a failed attempt, a canary arm that was not reading the fix it was supposed to be grading. Each was found because the loop keeps a record that can be checked against ground truth, each was fixed in production between runs, and each was re-measured on the same protocol [1]. A retrieval benchmark would have seen none of them, because none of them is about retrieval. A system that improves itself has to be inspectable in exactly this way, and the record it keeps for its own gates is what makes it so.

6 Related work

Agents that improve from experience. Self-Refine [7] and STaR [8] improve a model's outputs or its training data within a task or a training run. Reflexion [3] writes a verbal reflection after a failed trial for the next trial on the same task. ExpeL [4] distils cross-task insights from past trajectories into text the agent reads later; Voyager [5] keeps a library of verified skills; Agent Workflow Memory [6] induces reusable workflows from past successes. All of them write something after an episode that a later episode reads. None attributes later outcomes back to what was written, none detects that a written rule has stopped holding, and none tests a correction on the failing cases and on live traffic before every later episode reads it. Those three steps are the engine.

Self-modifying systems. Gödel machines [9], the Darwin Gödel Machine [10] and AlphaEvolve [11] change their own code under a proof or a benchmark gate. The gate is the shared idea. The object differs: the engine changes operating knowledge and the prompts and models that produce corrections to it, each change is measured against a live control arm, and each is reversible.

Release engineering and bandits. The engine's third gate is a canary release with a non-inferiority bound, in the tradition of controlled online experiments [12]. The act, explore and escalate recommendation that level 0 returns with each retrieval is a bandit over actions within a situation [13], with exploration assigned per agent; the engine's contribution is the attribution that feeds it, not the policy.

Benchmarks. τ-bench and τ^2-bench [14, 15] evaluate tool-using agents against fixed rules, and LongMemEval [16] evaluates recall over long histories. Neither changes a rule under the agent and grades what it does next, which is the measurement this paper rests on.

7 Limits

The evidence has limits, and some of them are findings about the approach.

The moved-half advantage over the context window is 4 points and does not approach significance ($p = 0.544$ paired). What separates the two systems is cost (3×) and what each can do for a second agent; this number does not. The kept-half loss to the context window (24 points, Holm 0.001) is the price of a selective, outcome-weighted record on a rule that did not move: the engine revises what the context window replays. Whether that trade is right depends on how often an operation's rules change, which is an empirical question about the operation.

Inside a 100-episode single-agent cell, the repair loop and the canary add nothing measurable to outcome learning. The loop's value is in the ten-agent curve, in what it refused, and in the one wrong correction it stopped; the single-agent aggregate does not show it, and we do not claim it does. In the eight confirmatory ten-agent cells nothing shipped at all, because the canary's rule needs more graded traffic per arm than 200 episodes across ten agents supplied. The four ships and the one failure come from an exploratory run on development seeds with the traffic concentrated on the changed classes, and three of the four ships were on a class that was already at ceiling. The ten-agent curve has no baseline fleet under its protocol, so it shows that the engine keeps learning over 200 episodes and not that a store without the engine would not.

The joiner and Managed Model results are development-seed probes: six agents on an unconditional change for the first, and rung 0 of a four-rung ladder, under a relaxed promotion gate, for the second. Their confirmatory versions are registered and not yet run.

The benchmark is two families of synthetic tasks with oracle grading and a rule change the authors wrote. It knows the truth before and after every change, which is what makes the measurement possible and what makes it an approximation of an operation whose truth nobody writes down. Horizons are 100 and 200 episodes. The engine's cost per episode is three times the model alone and 1.3–1.8 times a pgvector store, and a third of the context window.

8 Conclusion

An agent that runs for a long time needs more than a place to put what it learned. It needs a process that notices when what it learned has stopped being true, writes the correction, proves it on the cases that failed, exposes it to a little live traffic, and keeps or reverts it on the evidence. That process is recursive self-improvement in the sense this paper uses, and the measurements here are, to our knowledge, the first of an engine doing it in production code against baselines that only store and recall. The engine is ahead on the population the change created, level or ahead on what did not change, three times cheaper than the strongest baseline, and able to hand what it learned

to a new agent intact, or to a model the team owns at a twenty-third of the cost. It also records what it refused to ship, which is the part of a self-improving system that lets you trust the rest.

References

[1] SenseLab Research. Recursive self-improvement, gated by evidence: an engine that lets deployed agents propose, test, canary and ship corrections to their own operating knowledge, measured where the world changes under them. Preprint v0.3, October 2026. Companion paper; preregistration `HYPOTHESIS_RSI.md`, analysis tag `analysis-freeze-rsi`.

[2] SenseLab Research. From memory to recursive self-improvement: agents that learn from outcomes, agents that only remember, and a model trained on what they learned, when the tasks never repeat and the rules change. Preprint, 2026.

[3] N. Shinn, F. Cassano, A. Gopinath, K. Narasimhan, S. Yao. Reflexion: Language agents with verbal reinforcement learning. NeurIPS 2023.

[4] A. Zhao, D. Huang, Q. Xu, M. Lin, Y.-J. Liu, G. Huang. ExpeL: LLM agents are experiential learners. AAAI 2024 (arXiv:2308.10144).

[5] G. Wang et al. Voyager: An open-ended embodied agent with large language models. TMLR 2024 (arXiv:2305.16291).

[6] Z. Z. Wang, J. Mao, D. Fried, G. Neubig. Agent Workflow Memory. arXiv:2409.07429, 2024.

[7] A. Madaan et al. Self-Refine: Iterative refinement with self-feedback. NeurIPS 2023.

[8] E. Zelikman, Y. Wu, J. Mu, N. D. Goodman. STaR: Bootstrapping reasoning with reasoning. NeurIPS 2022.

[9] J. Schmidhuber. Gödel machines: Fully self-referential optimal universal self-improvers. In *Artificial General Intelligence*, Springer, 2007.

[10] J. Zhang, S. Hu, C. Lu, R. Lange, J. Clune. Darwin Gödel Machine: Open-ended evolution of self-improving agents. arXiv:2505.22954, 2025.

[11] A. Novikov et al. AlphaEvolve: A coding agent for scientific and algorithmic discovery. arXiv:2506.13131, 2025.

[12] R. Kohavi, D. Tang, Y. Xu. *Trustworthy Online Controlled Experiments: A Practical Guide to A/B Testing*. Cambridge University Press, 2020.

[13] T. Lattimore, C. Szepesvári. *Bandit Algorithms*. Cambridge University Press, 2020.

[14] S. Yao, N. Shinn, P. Razavi, K. Narasimhan. τ-bench: A benchmark for tool-agent-user interaction in real-world domains. arXiv:2406.12045, 2024.

[15] V. Barres et al. τ^2-bench: Evaluating conversational agents in a dual-control environment. arXiv:2506.07982, 2025.

[16] D. Wu et al. LongMemEval: Benchmarking chat assistants on long-term interactive memory. ICLR 2025.